

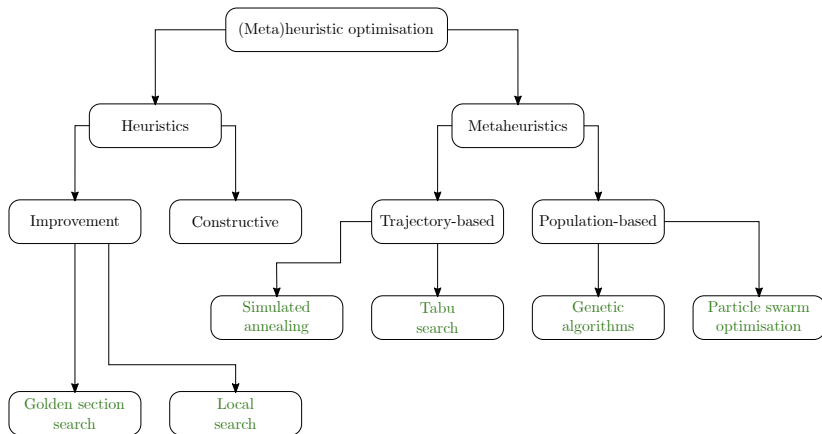
SIMULATED ANNEALING

Lecture 24: Presented by Jacomine Grobler



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(Meta)heuristics



The physical process of *annealing* in metallurgy:

- Heating and subsequently cooling a material in stages, called *epochs*
- Heating allows for easier formation of the metal
- Iterative cooling results in strengthened material

Simulated annealing in optimisation:

- Similar to local search
- A neighbouring solution is selected randomly
- If the objective function value improves, it is accepted
- If it does not improve, it accepted with a probability proportional to:
 - The degree of objective function value degradation
 - A temperature

The Metropolis criterion

The solution perturbation $\bar{\mathbf{x}}^k$ is accepted with probability

$$P = \begin{cases} 1 & \text{if } f(\bar{\mathbf{x}}^k) < f(\mathbf{x}^k) \\ \exp\left(\frac{f(\mathbf{x}^k) - f(\bar{\mathbf{x}}^k)}{T^i}\right) & \text{otherwise} \end{cases}$$

in the case of a minimisation problem, or with probability

$$P = \begin{cases} 1 & \text{if } f(\bar{\mathbf{x}}^k) > f(\mathbf{x}^k) \\ \exp\left(\frac{f(\bar{\mathbf{x}}^k) - f(\mathbf{x}^k)}{T^i}\right) & \text{otherwise} \end{cases}$$

in the case of a maximisation problem according to the Metropolis criterion, where in both cases T^i denotes the temperature during epoch i , within which search iteration k occurs. If a worsening solution $\bar{\mathbf{x}}^k$ is not accepted, we say that it is *rejected*, in which case the new current solution $\mathbf{x}^{k+1}$, to be considered during the next search iteration, remains $\mathbf{x}^k$.

Choices to be performed empirically based on a problem-by-problem basis:

- Cooling schedule
- Reheating schedule
- Initial temperature
- Stopping criteria

- Linear cooling: $T^{i+1} = T^i - \alpha$
- Geometric cooling: $T^{i+1} = \beta T^i$
- Logarithmic cooling: $T^{i+1} = \gamma / \log(\delta + i)$
- Adaptive cooling schedule of Huang *et al.* :
$$T^{i+1} = T^i \exp\left(-\frac{\lambda T^i}{\sigma^i}\right)$$
- Adaptive cooling schedule of Van Laarhoven and Aarts:
$$T^{i+1} = \frac{T^i}{1 + \frac{\ln(1+\xi)}{3\sigma^i} T^i}$$

Where σ^i denotes the standard deviation of the objective function values of search points encountered up to the end of epoch i .

The parameters α , β , γ , δ and ξ above are constants and are all referred to as cooling parameters

- Linear reheating: $T^{i+1} = T^i + \alpha'$
- Geometric reheating: $T^{i+1} = \beta' T^i$

Initial temperature

It should be relatively likely to accept worsening solutions at the start of the search, in a bid to promote exploration of the search space initially.

A popular rule of thumb involves carrying out a pilot search during which all worsening perturbations are accepted.

The initial temperature is then taken as

$$T^0 = \frac{\overline{\Delta E}^+}{\ln \chi},$$

where $\overline{\Delta E}^+$ denotes the average change in objective function values observed during the pilot search, computed over all worsening perturbations carried out, and where χ denotes the ideal so-called *acceptance ratio* (the ratio of the number of worsening perturbations carried out to the total number of perturbations carried out during the pilot search).

- Stopping after a specified number of search iterations
- Stopping once a specified number of reheatings have occurred
- Stopping when the incumbent solution no longer improves

Simulated annealing

Algorithm 4.5 (Simulated annealing metaheuristic (for minimisation))

Input: A multivariate function $f : \mathbb{R}^n \mapsto \mathbb{R}$, an initial estimate $\mathbf{x}^0$ of a local optimum of f , an initial temperature T , a cooling schedule, a reheating schedule, a maximum number $a_{\max}$ of solution perturbations that may be accepted per epoch, a maximum number $r_{\max}$ of solution perturbations that may be rejected per epoch, a maximum number $k_{\max}$ of search iterations that may be performed in total, and a maximum number $p_{\max}$ of epochs that may elapse in total

Output: An approximation of a local optimum of the function f

Step 1 $k \leftarrow 0, i \leftarrow 0, p \leftarrow 0, a \leftarrow 0, r \leftarrow 0, \text{Incumbent} \leftarrow \mathbf{x}^0$

Step 2 If $k = k_{\max}$ or $p = p_{\max}$, then output Incumbent and Stop

Step 3 If $r = r_{\max}$, then

Step 4 $i \leftarrow i + 1, p \leftarrow p + 1, a \leftarrow 0, r \leftarrow 0$

Step 5 Increase T according to the reheating schedule and Go to Step 2

Step 6 If $a = a_{\max}$, then

Step 7 $i \leftarrow i + 1, a \leftarrow 0, r \leftarrow 0$

Step 8 Decrease T according to the cooling schedule and Go to Step 2

Step 9 Generate a perturbation $\mathbf{x}^k \rightsquigarrow \bar{\mathbf{x}}^k$ stochastically

Step 10 If $\text{rand}(0, 1) < \min\{1, \exp((f(\mathbf{x}^k) - f(\bar{\mathbf{x}}^k))/T)\}$ then

Step 11 $\mathbf{x}^{k+1} \leftarrow \bar{\mathbf{x}}^k, a \leftarrow a + 1$

Step 12 If $f(\mathbf{x}^{k+1}) < f(\text{Incumbent})$ then Incumbent $\leftarrow \mathbf{x}^{k+1}$

Step 13 $k \leftarrow k + 1$ and Go to Step 6

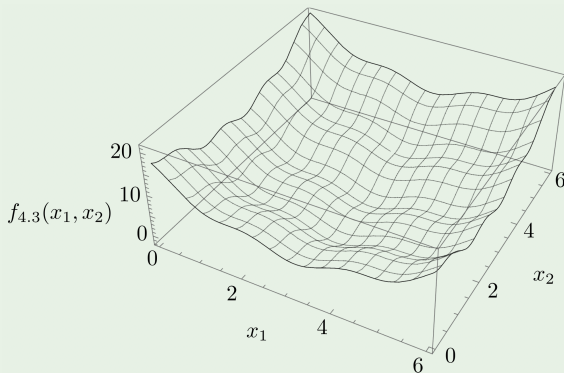
Step 14 Else $\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k, r \leftarrow r + 1, k \leftarrow k + 1$, Go to Step 3

Example 4.6 (pp. 133–135)

Minimise the function

$$f_{4.3}(x_1, x_2) = (x_1 - 3.14)^2 + (x_2 - 2.72)^2 + \sin(3x_1 + 1.41) + \sin(4x_2 - 1.73)$$

approximately on the domain $(x_1, x_2) \in [0, 6] \times [0, 6]$ by applying the method of simulated annealing, starting out from the initial estimate $\mathbf{x}^0 = [1 \ 1]^T$.



Example 4.6 (pp. 133–135)

We adopt the stochastic neighbourhood perturbation operator

$$\begin{bmatrix} x_1^k \\ x_2^k \end{bmatrix} \rightsquigarrow \begin{cases} [x_1^k + \varepsilon & x_2^k]^T & \text{if } r^1 \in [0, \frac{1}{4}] \\ [x_1^k & x_2^k + \varepsilon]^T & \text{if } r^1 \in (\frac{1}{4}, \frac{1}{2}] \\ [x_1^k - \varepsilon & x_2^k]^T & \text{if } r^1 \in (\frac{1}{2}, \frac{3}{4}] \\ [x_1^k & x_2^k - \varepsilon]^T & \text{if } r^1 \in (\frac{3}{4}, 1] \end{cases}$$

where $\varepsilon > 0$ is a step length and r^1 is a random number drawn from the unit interval $[0, 1]$ according to a uniform distribution.

Adopt both a geometric reheating schedule and a geometric cooling schedule with heating factor $\beta' = 1.1$ and cooling factor $\beta = 0.9$. Take the initial temperature as $T^0 = 1$, and terminate a search epoch either if $a_{\max} = 10$ solutions have been accepted during a single epoch or if $r_{\max} = 4$ solutions have been rejected during a single epoch. Terminate the entire search if either $k_{\max} = 300$ search iterations or $p_{\max} = 10$ reheatings have occurred (whichever occurs first).

Example 4.6 (pp. 133–135)

